
EDUCATION

RV Institute of Technology

Bachelor's of Engineering: Computer Science

Core: Operating Systems, DBMS, DSA, Computer Networks.

Chaithanya Layout, 8th Phase, J.P

2023-2027

CGPA-8.23

SKILLS

Languages: Python (proficient) (900+ LeetCode), Java.

Backend & APIs: FastAPI, REST, Pydantic.

Cloud & DevOps: Docker, AWS (EC2, S3, Lambda), CI/CD.

Machine Learning & AI: Deep Learning, scikit-learn, TensorFlow, Keras, OpenCV.

Data & Tools: Pandas, NumPy, Matplotlib, Seaborn

Development: Git/GitHub.

PROJECTS

Driver Drowsiness Detection System | [Github](#)

[Working Video](#)

Built a **real-time driver drowsiness detection system** using CNN to analyze eye state, facial alertness from live video.

- Designed and evaluated **15+ CNN architectures**, achieving **98.7% eye-state accuracy** and **98.4% face-state accuracy** on curated datasets.
- Engineered a **low-latency video inference pipeline** with frame capture, prediction smoothing, and alert triggering.
- Built **modular detection components** (face detection, eye inference, alert logic) enabling independent testing & tuning
- Implemented a **threshold-based alert mechanism with user acknowledgment**, reducing false positives in real-world.

Insight Weaver – Scientific Discovery Copilot | [Github](#)

[Working Video](#)

Built an evidence-first GraphRAG pipeline: scientific PDFs → entity graph → hybrid retrieval → grounded Gemma reasoning.

- Engineered section-aware chunking, scientific entity extraction, BM25 + vector retrieval fusion, and graph-context expansion for cross-paper reasoning.
- Developed contradiction detection and entity-grounded hypothesis generation over multi-paper scientific literature using locally hosted Gemma 4 4B with automatic 2B fallback inference.
- Dockerized FastAPI-based retrieval and reasoning pipelines for reproducible local LLM deployment and low-latency inference workflows.

TACOS — Real-Time Toxic Comment Moderation System | [Github](#)

[Live Demo](#)

Real-time AI moderation system to detect, delete, and escalate toxic social media comments.

- Trained a DistilBERT on 2M+ comments to detect six toxicity categories; achieved **Micro-F1: 0.75, Macro-F1: 0.54**.
- Designed inference pipeline using **FastAPI**, handling real-time requests with sub-second response latency.
- Converted model outputs into actionable moderation decisions (allow / warn / block) using threshold-based logic.
- Built a Streamlit interface to inspect predictions and confidence scores for each class.
- Deployed a containerized application on **Hugging Face Spaces** using Docker.
- Structured code into separate modules for model inference, decision logic, and UI to simplify testing and iteration.

Achievements

- Ranked #1693 / 82,790** – [Amazon ML Hackathon 2025](#) (Product Price Prediction).
- 900+ LeetCode challenges** solved.
- Semifinalist, RIFT '26 – 24 Hour Hackathon** by PhysicsWallah.
- Semifinalist, Adivya 2.0 at IIT Ropar**, “[SkillQuest AI](#)”, gamified AI learning platform, adaptive assessments, tutoring.
- Ranked 3593 among 32K+ participants** in [CodeWar \(IIT Ropar\)](#), a national-level CodeChef contest.
- Developed 25+ projects** involving AI/ML, REST and Dockerized deployments over AWS.

Community & Leadership

- U&I Volunteer Teacher**—mentoring underprivileged and orphaned students through weekly academic support sessions.
- Led **fundraising initiatives** and **volunteer teams** to support students' educational expenses and learning resources.
- NSS Coordinator**— at National Service Scheme, organizing workshops and coordinating volunteers, logistics, and campus events.